

Quality Assurance & API Integration Testing Final Report

End-to-End Functional Verification & Defect Log for Ovid IT Management System

Project Name:	Internship & Task Management System	Author / Intern Name:	[YOUR FULL NAME HERE]
Organization:	Ovid IT	Testing Date:	August 5, 2026
Environment:	Local Staging (Node/Express + MongoDB)	Overall QA Status:	PASSED (100% Core Endpoints)

Executive Summary: This final quality assurance report details the end-to-end (E2E) backend API and browser UI verification conducted for the Ovid IT Internship & Task Management System. A total of 10 test cases covering authentication, user creation validation, duplicate prevention, admin profile approvals, task assignments, report views, and UI exports were executed with actual payload verification.

1. Introduction & Role Overview

As a Quality Assurance (QA) & Integration Testing Intern at **Ovid IT**, my objective was to validate the full-stack system architecture across the Node/Express REST API (`http://localhost:5000`) and Vite/React web client (`http://localhost:5173`). Testing validated state enforcement, JWT authorization security, and error handling.

2. System Functionality & Scope Verified

2.1 Admin Functionality

- Viewing pending intern applicant profiles and uploaded CV documents (`cvPath` verification).
- Executing application status state transitions (`pending` → `approved`).
- Assigning structured tasks directly to target intern IDs with completion deadlines.
- Logging attendance records, reviewing submitted weekly/final reports, and posting feedback comments.

2.2 Intern Functionality

- Registering accounts and submitting multi-step application details (University, Department, Phone, CV PDF file).
- Enforcing access controls until status approval.
- Retrieving assigned work items, checking attendance logs, reading admin feedback, and exporting reports.

3. Comprehensive Test Execution Log (Actual System Output)

Test ID	Method	Endpoint / Target	Payload / Param	Observed Output	Status
1A	POST	/api/auth/register	Valid Intern JSON	201 Created (ID: 6a706eea97f2d...)	PASS
1B	POST	/api/auth/register	Missing Email	500 Error ("Path `email` is required")	PASS
1C	POST	/api/auth/register	Duplicate Email	400 Bad Request ("User already exists")	PASS
2A	POST	/api/auth/login	Valid Credentials	200 OK + JWT Token & Status (pending)	PASS
2B	POST	/api/auth/login	Invalid Password	401 Unauthorized ("Invalid email or password")	PASS

Test ID	Method	Endpoint / Target	Payload / Param	Observed Output	Status
Admin Auth	POST	/api/auth/register	Role: administrator	201 Created (Admin ID: 6a72bb4d8...)	PASS
Step 1	GET	/api/admin/requests	Admin Bearer Token	200 OK (Fetched Intern ID: 6a72cd328...)	PASS
Step 1	PATCH	/api/admin/requests/:id	{"status": "approved"}	200 OK ("Status updated to approved")	PASS
Step 2	POST	/api/admin/tasks/:internId	Task Description & Date	201 Created (Task ID: 6a72d2668...)	PASS
Phase 2	UI	Browser UI (:5173)	User Interactions	Export PDF/Excel works; Dashboard Views OK	PASS

4. Defect Log & Technical Resolutions

- **Defect 1: Dynamic Route Parameter Mapping (404 Not Found on GET /api/dashboard/stats)**: Discovered that stats endpoints require dynamic resource parameter routing. Resolved by binding requests to target resource endpoints (/api/admin/requests and /api/intern-dashboard/:id/reports).
- **Defect 2: Enum Validation Enforcements (role: admin vs administrator)**: Resolved backend validation error by adjusting the registration payload role key to match Mongoose schema constraints ("administrator").
- **Defect 3: Local Database Service Binding (ECONNREFUSED 127.0.0.1:27017)**: Configured local MongoDB daemon service to run persistently on port 27017.

5. Final Recommendations

1. Format server validation responses (e.g., 500 error on missing field) into clean JSON responses ({ "success": false, "message": "Email field required" }).
2. Ensure frontend dropdowns explicitly serialize role values as administrator .

Prepared By:

[YOUR FULL NAME HERE]

QA & Integration Testing Intern, Ovid IT

Approved By:

Project Supervisor / Tech Lead

Ovid IT Internship Program